Physical Properties of Fluids and Flow Characteristics of Valves, Fittings, and Pipe

APPENDIX A

The physical properties of many commonly used fluids are required for the solution of flow problems. This appendix presents a compilation of such properties obtained from various reference sources. In those cases where the information given by the source reference is in Imperial units this has been converted and is presented here in terms of SI units.

Most texts on the subject of fluid mechanics cover in detail the flow through pipe, but the flow characteristics of valves and fittings are given little, if any, attention, probably because the information has not been available. This appendix includes a presentation of data which provides a basis for calculating the resistance coefficient 'K' for various types of valves and fittings and the method of employing this coefficient to obtain pressure drop or head loss through valves and fittings is explained in Chapter 2.

The Y net expansion factors for discharge of compressible fluids from piping systems, which are presented here, provide means for a greatly simplified solution of a heretofore complex problem.

Viscosity of Water and Steam[7] – in centipoise (μ)

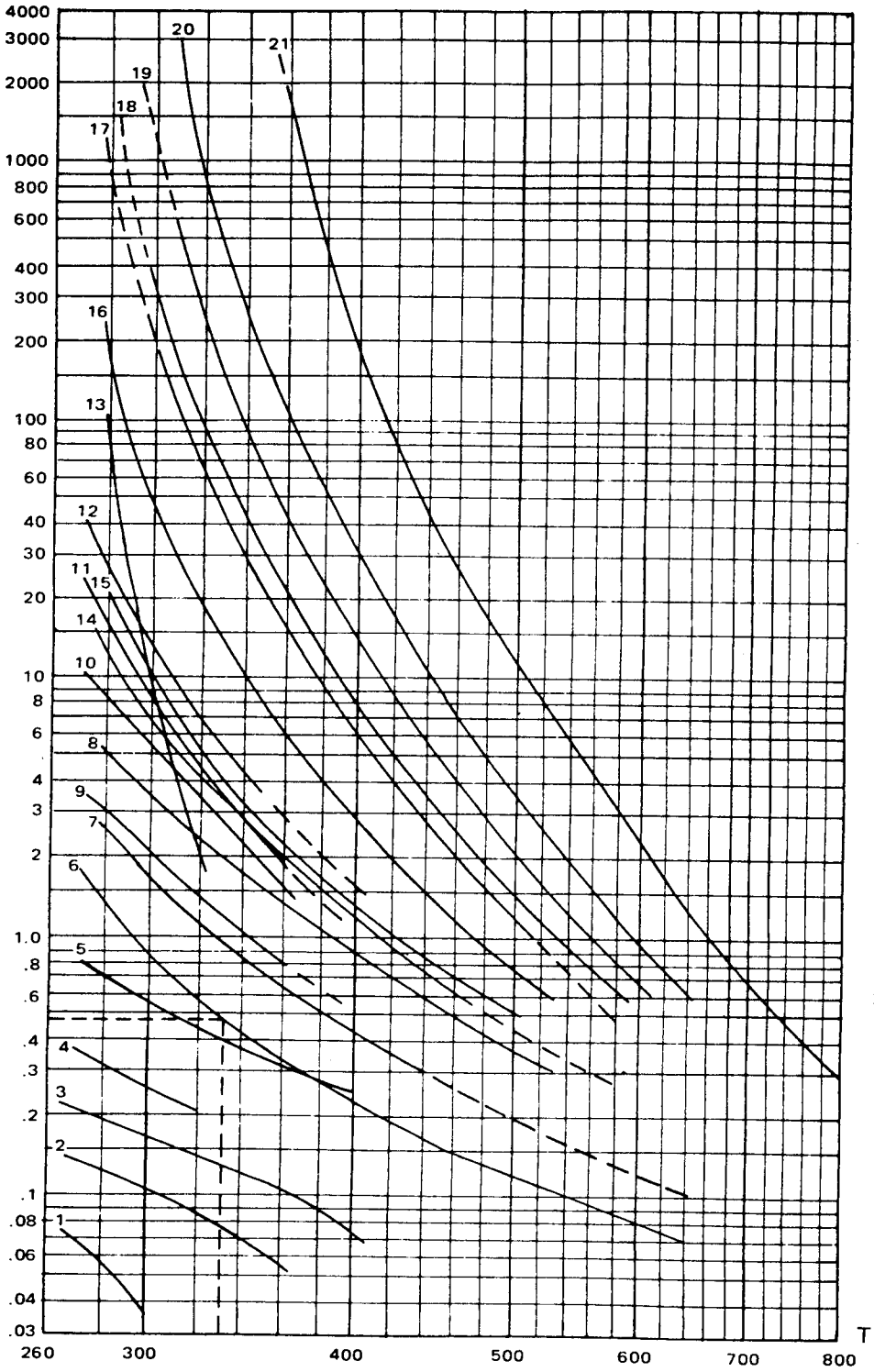
Temp. °C	Pressure. Bar Absolute														
	1	5	10	25	50	75	100	150	200	300	400	500	600	700	800
0	1.750	1.750	1.750	1.750	1.750	1.750	1.750	1.740	1.740	1.740	1.730	1.720	1.720	1.710	1.710
50	.544	.544	.544	.544	.545	.545	.545	.546	.546	.547	.548	.549	.550	.551	.552
100	.012	.279	.279	.280	.280	.280	.281	.282	.283	.285	.287	.289	.291	.293	.295
150	.014	.181	.181	.182	.182	.183	.183	.184	.186	.188	.190	.192	.194	.197	.199
200	.016	.016	.016	.134	.135	.135	.136	.137	.138	.140	.143	.145	.148	.150	.152
250	.018	.018	.018	.018	.107	.108	.108	.110	.111	.113	.116	.118	.121	.123	.126
300	.020	.020	.020	.020	.020	.020	.090	.092	.093	.095	.098	.101	.103	.106	.108
350	.022								.073	.078	.082	.085	.087	.089	.091
375	.023	.023	.023	.024	.024	.024	.025	.026	.029⊙	.066	.072	.076	.079	.082	.085
400	.024	.024	.024	.025	.025	.025	.026	.027	.029	.046	.063	.069	.074	.077	.080
425	.025	.025	.025	.026	.026	.026	.027	.028	.029	.034	.050	.061	.067	.071	.075
450	.026	.026	.026	.027	.027	.027	.028	.028	.030	.033	.041	.052	.060	.065	.069
475	.027	.027	.027	.028	.028	.028	.029	.029	.030	.033	.038	.046	.053	.060	.064
500	.028	.028	.028	.029	.029	.029	.029	.030	.031	.033	.037	.042	.048	.054	.060
550	.030	.030	.030	.031	.031	.031	.031	.032	.033	.035	.037	.040	.044	.048	.053
600	.032	.032	.033	.033	.033	.033	.033	.034	.034	.036	.038	.040	.043	.046	.049
650	.034	.034	.035	.035	.035	.035	.035	.036	.036	.038	.039	.041	.043	.045	.048
700	.036	.037	.037	.037	.037	.037	.037	.038	.038	.039	.041	.042	.044	.046	.048

Notes. (1) The entry shown for 0°C and 1 bar relates to a metastable liquid state. The stable state is here solid.
(2) ⊙ Critical point, 374.15°C, 221.2 bar.

Source of data: NEL Steam Tables 1964 (HMSO, Edinburgh)[7]

Viscosity of Water and Liquid Petroleum Products

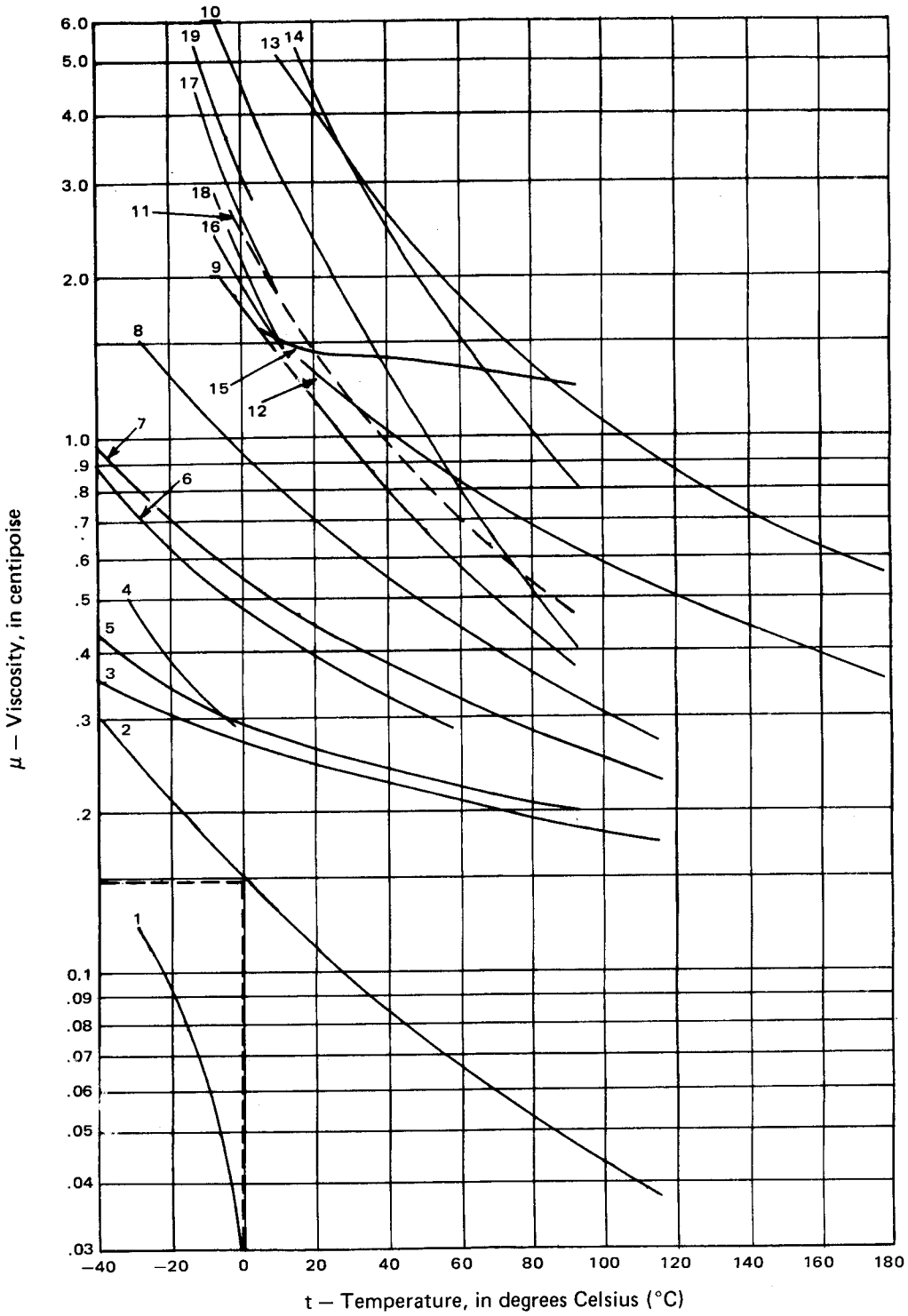
Adapted from data extracted from
Bibliography references 8, 12 and 23



1. Ethane (C_2H_6)
2. Propane (C_3H_8)
3. Butane (C_4H_{10})
4. Natural Gasoline
5. Gasoline
6. Water
7. Kerosene
8. Distillate
9. 48 Deg. API Crude
10. 40 Deg. API Crude
11. 35.6 Deg. API Crude
12. 32.6 Deg. API Crude
13. Salt Creek Crude
14. Fuel 3 (Max.)
15. Fuel 5 (Min.)
16. SAE 10 Lube (100 V.I.)
17. SAE 30 Lube (100 V.I.)
18. Fuel 5 (Max.) or Fuel 6 (Min.)
19. SAE 70 Lube (100 V.I.)
20. Bunker C Fuel (Max.) and M.C. Residuum
21. Asphalt

Example: Find the viscosity of water at 60°C
Solution: 60°C = 273 + 60 = 333 K
Viscosity of water at 333 K = 0.47 centipoise (curve 6)

Viscosity of Various Liquids



1. Carbon Dioxide . . CO_2
2. Ammonia . . . . . NH_3
3. Methyl Chloride . CH_3Cl
4. Sulfur Dioxide . SO_2
5. Freon 12 . . . . . F-12
6. Freon 114 . . . . . F-114
7. Freon 11 . . . . . F-11
8. Freon 113 . . . . . F-113
9. Ethyl Alcohol
10. Isopropyl Alcohol
11. 20% Sulfuric Acid . . . . . 20% H_2SO_4
12. Dowtherm E
13. Dowtherm A
14. 20% Sodium Hydroxide . . 20% NaOH
15. Mercury
16. 10% Sodium Chloride Brine . . . 10% NaCl
17. 20% Sodium Chloride Brine . . . 20% NaCl
18. 10% Calcium Chloride Brine. . . 10% $CaCl_2$
19. 20% Calcium Chloride Brine. . . 20% $CaCl_2$

Example: The viscosity of ammonia at 0°C is 0.15 centipoise.

Adapated from data extracted from Bibliography references 5, 8, 11

Viscosity of Gases

The curves for hydrocarbon vapo rs and natural gases in the chart at the upper right are adapted from data taken from Maxwell[15]; the curves for all other gases (except helium[27]) in the chart are based upon Sutherland's formula, as follows:

$$\mu = \mu_0 \left(\frac{T_0 + C}{T + C} \right) \left(\frac{T}{T_0} \right)^{3/2}$$

where:

μ = viscosity, in centipoise at temperature T.

μ_0 = viscosity, in centipoise at temperature T_0.

T = absolute temperature, in Kelvin (273 + °C), for which viscosity is required.

T_0 = absolute temperature, in Kelvin, for which viscosity is known.

C = Sutherland's constant.

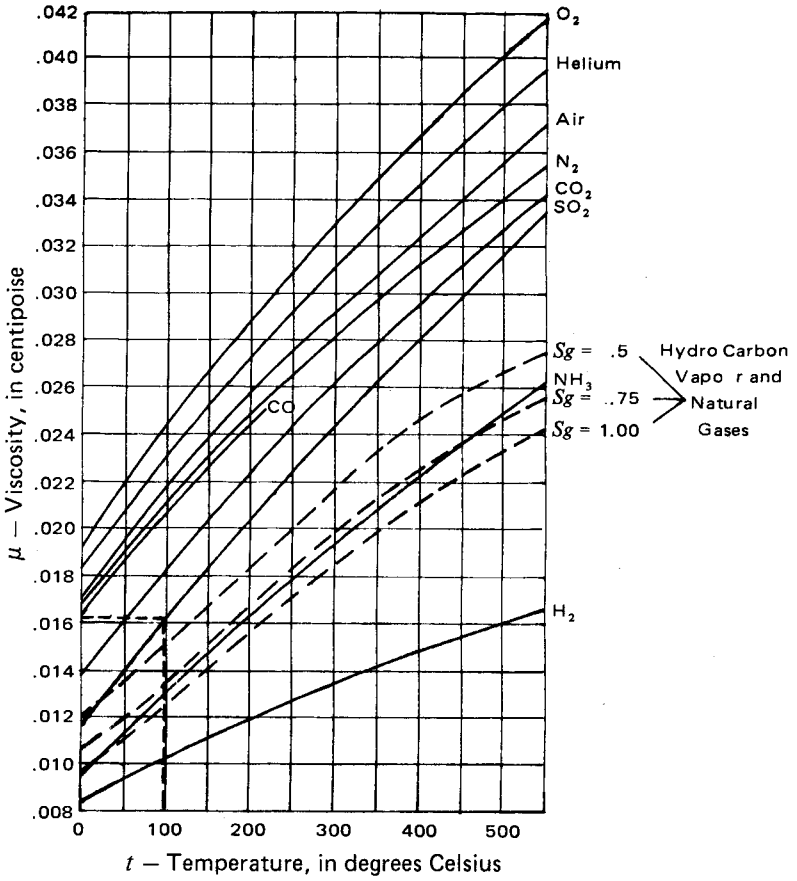
Note: The variation of viscosity with pressure is small for most gases. For gases given on this page, the correction of viscosity for pressure is less than 10 percent for pressures up to 35 bar.

Fluid	Approximate Values of "C"
O_2	127
Air	120
N_2	111
CO_2	240
CO	118
SO_2	416
NH_3	370
H_2	72

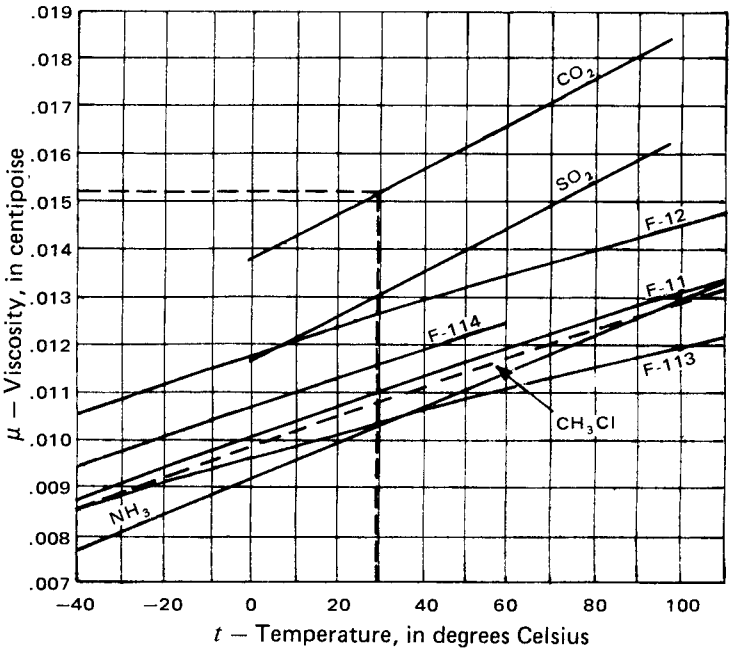
Upper chart example: The viscosity of sulfur dioxide gas (SO_2) at 100°C is 0.0162 centipoise.

Lower chart example: The viscosity of carbon dioxide gas (CO_2) at about 30°C is 0.0152 centipoise.

Viscosity of Various Gases



Viscosity of Refrigerant Vapo rs
(saturated and superheated vapo rs)



Physical Properties of Water

Temperature of Water t Degrees Celsius	Saturation Pressure p' Bar Absolute	Specific Volume $\bar{V} \times 10^3$ Cubic Decimeters per Kilogram	Density ρ Kilograms per Cubic Meters
.01	.006112	1.0002	999.8
5	.008719	1.0001	999.9
10	.012271	1.0003	999.7
15	.017041	1.0010	999.0
20	.023368	1.0018	998.2
25	.031663	1.0030	997.0
30	.042418	1.0044	995.6
35	.056217	1.0060	994.0
40	.073750	1.0079	992.2
45	.09582	1.0099	990.2
50	.12335	1.0121	988.0
55	.15740	1.0145	985.7
60	.19919	1.0171	983.2
65	.25008	1.0199	980.5
70	.31160	1.0228	977.7
75	.38547	1.0258	974.8
80	.47359	1.0290	971.8
85	.57803	1.0324	968.6
90	.70109	1.0359	965.3
95	.84526	1.0396	961.9
100	1.01325	1.0435	958.3
110	1.4326	1.0515	951.0
120	1.9853	1.0603	943.1
130	2.7012	1.0697	934.8
140	3.6136	1.0798	926.1
150	4.7597	1.0906	916.9
160	6.1805	1.1021	907.4
170	7.9203	1.1144	897.3
180	10.0271	1.1275	886.9
190	12.552	1.1415	876.0
200	15.551	1.1565	864.7
225	25.504	1.1992	833.9
250	39.776	1.2512	799.2
275	59.49	1.3168	759.4
300	85.92	1.4036	712.5
325	120.57	1.5289	654.1
350	165.37	1.741	574.4
374.15	221.20	3.170	315.5

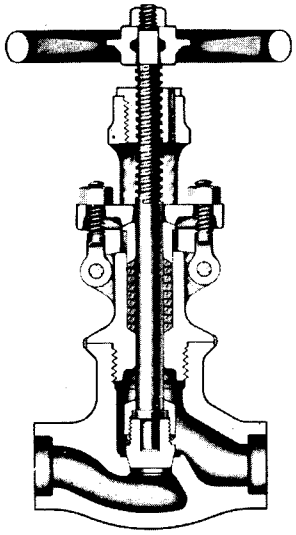
To convert Specific Volume from cubic decimeters per kilogram (dm^3/kg) to cubic meters per kilogram (m^3/kg) divide values in table by 10^3.

To convert Density from kilograms per cubic meter (kg/m^3) to kilograms per liter (kg/liter) divide values in table by 10^3.

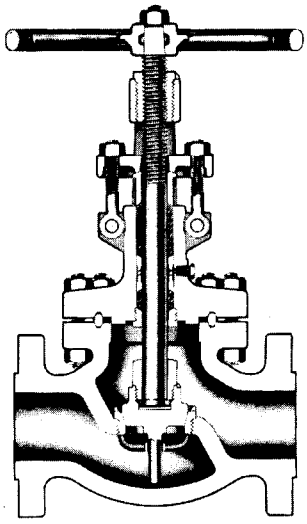
Specific gravity of water at 15°C = 1.00.

Data on pressure and volume abstracted from UK National Engineering Laboratory "Steam Tables 1964" with permission of HMSO.

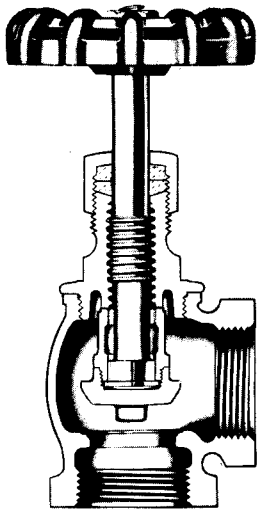
Types of Valves



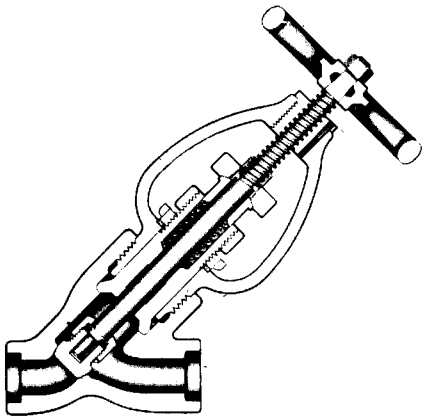
Conventional Globe Valve



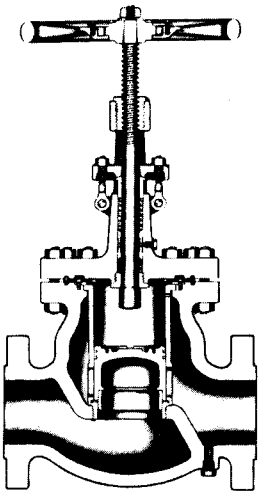
Conventional Globe Valve
With Disc Guide



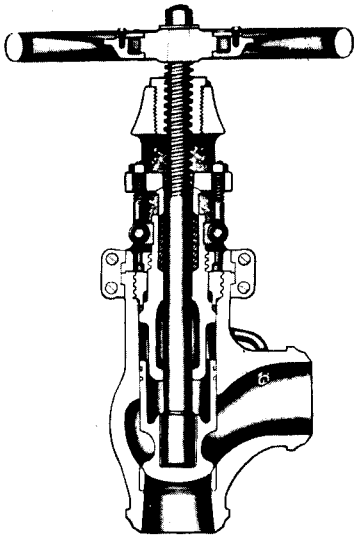
Conventional Angle Valve



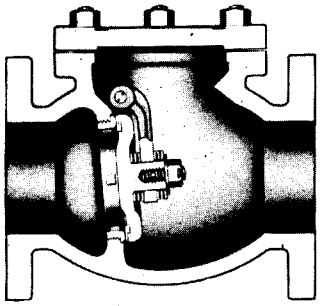
Y-Pattern Globe Valve
With Stem 45 degrees from Run



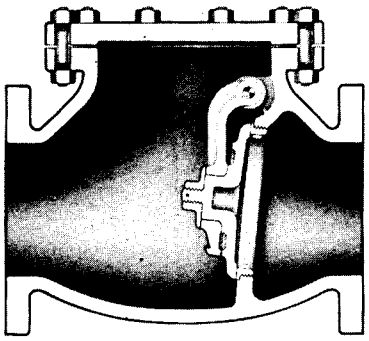
Globe Stop-Check Valve



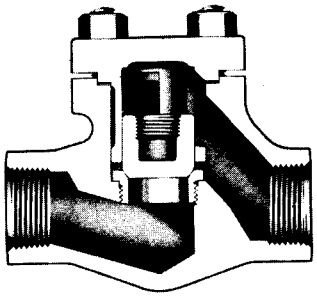
Angle Stop-Check Valve



Conventional Swing Check Valve

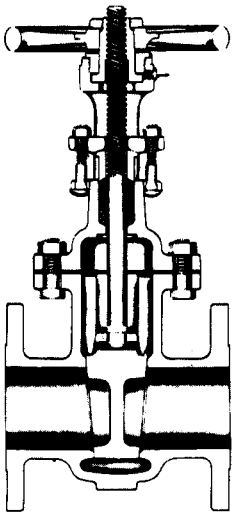


Clearway Swing Check Valve

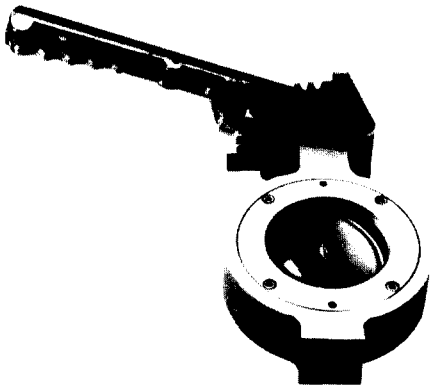


Globe Type Lift Check Valve

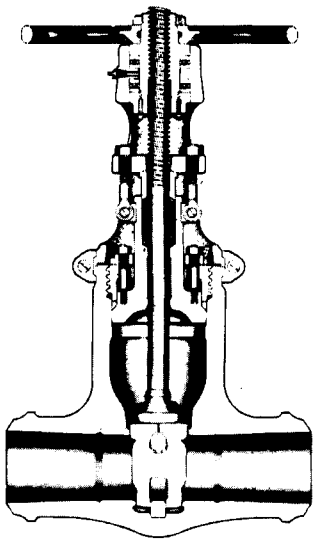
Types of Valves



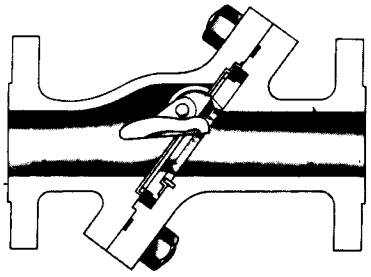
Wedge Gate Valve
(Bolted Bonnet)



High Performance Butterfly Valve



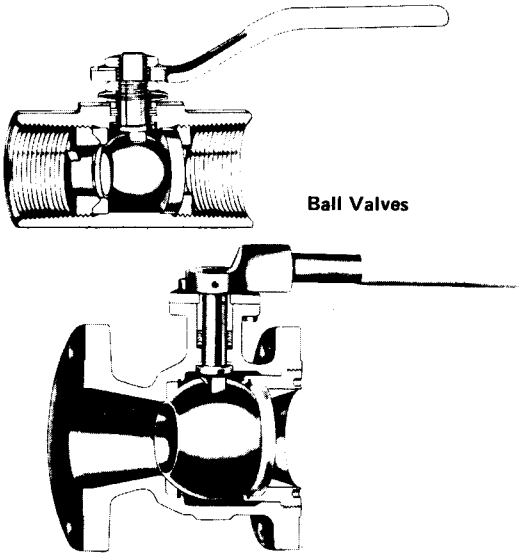
Flexible Wedge Gate Valve
(Pressure-Seal Bonnet)



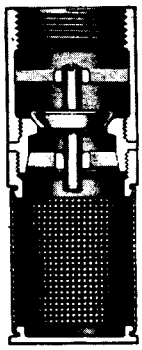
Tilting Disc Check Valve

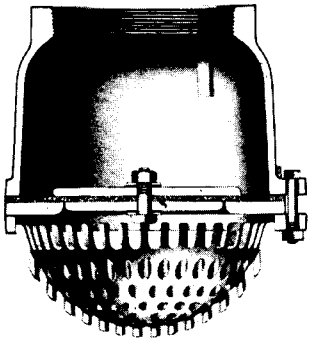


Butterfly Wafer Valve

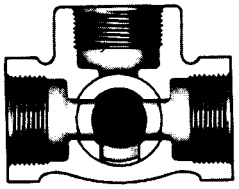


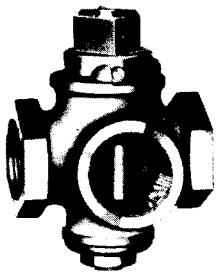
Ball Valves





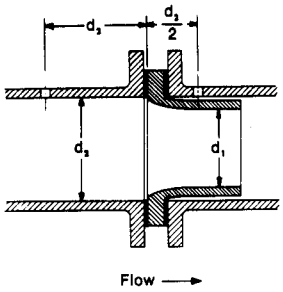
Foot Valves
Poppet and Hinged Types





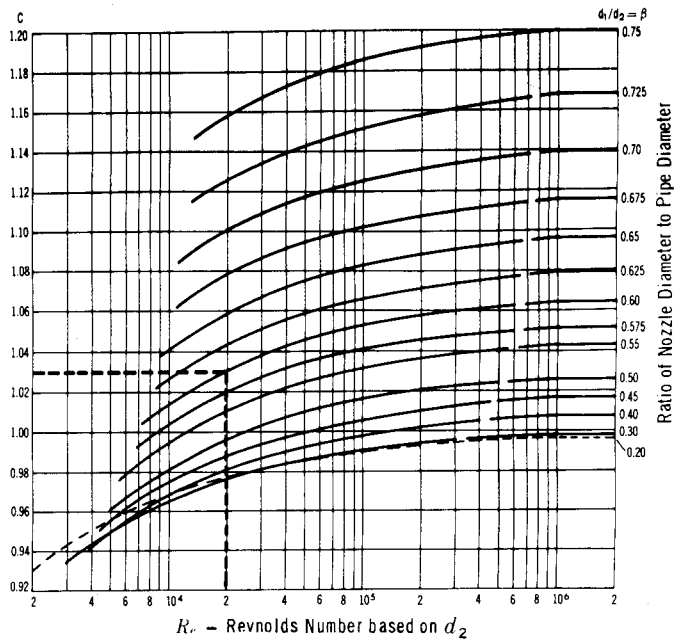
Three-Way Cock
Sectional and Outside Views

Flow Coefficient C for Nozzles[9]

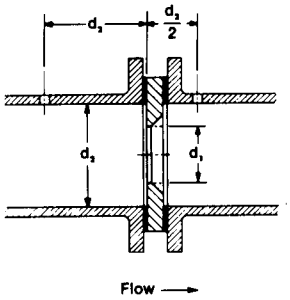


$$C = \frac{C_d}{\sqrt{1-\beta^4}}$$

Example: The flow coefficient C for a diameter ratio β of 0.60 at a Reynolds number of 20 000 (2×10^4) equals 1.03.

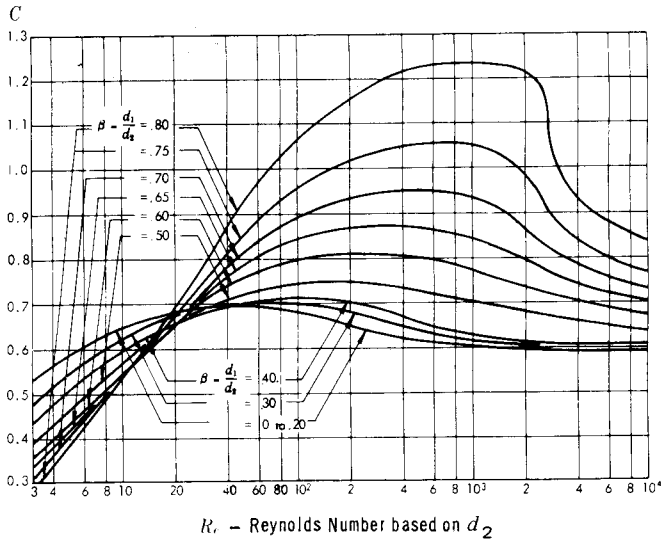


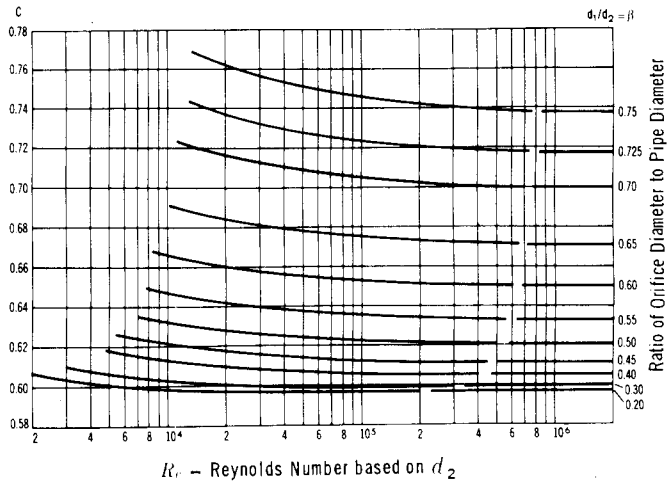
Flow Coefficient C for Square-Edge Orifices[9,17]



$$C = \frac{C_d}{\sqrt{1-\beta^4}}$$

$$K_{\text{orifice}} \simeq \frac{1-\beta^2}{C^2\beta^4}$$



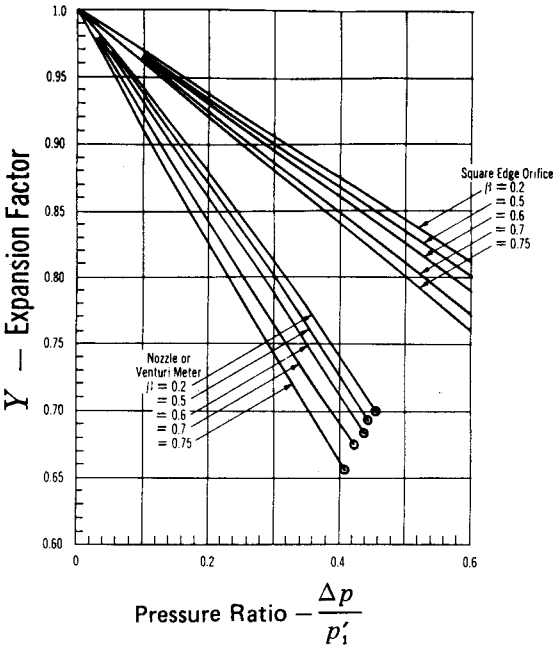


Net Expansion Factor, Y

For Compressible Flow through Nozzles and Orifices[9]

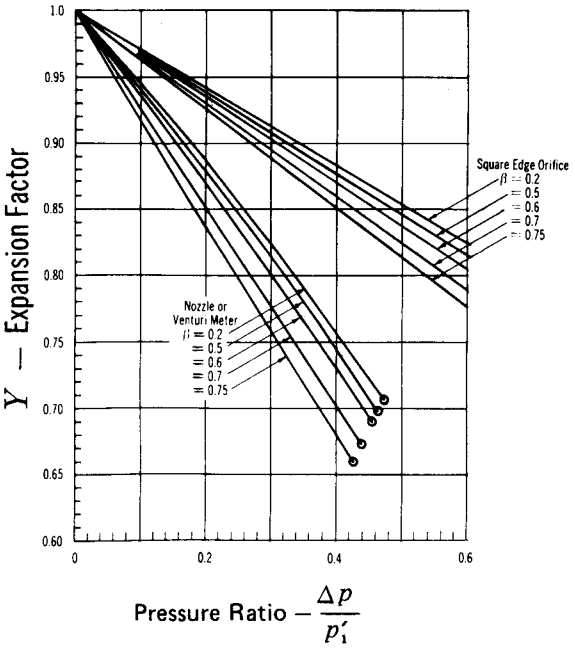
$\gamma = 1.3$ approximately

(for CO_2, SO_2, H_2O, H_2S, NH_3, N_2O, Cl_2, CH_4, C_2H_2, and C_2H_4)



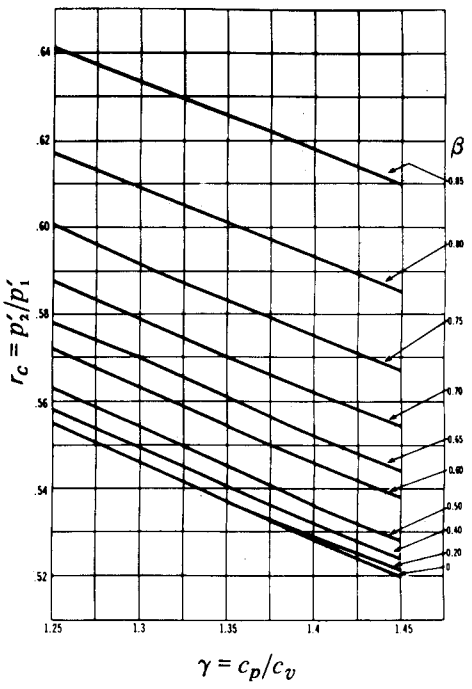
$\gamma = 1.4$ approximately

(for Air, H_2, O_2, N_2, CO, NO, and HCl)



Critical Pressure Ratio, r_c

For Compressible Flow through Nozzles and Venturi Tubes[9]

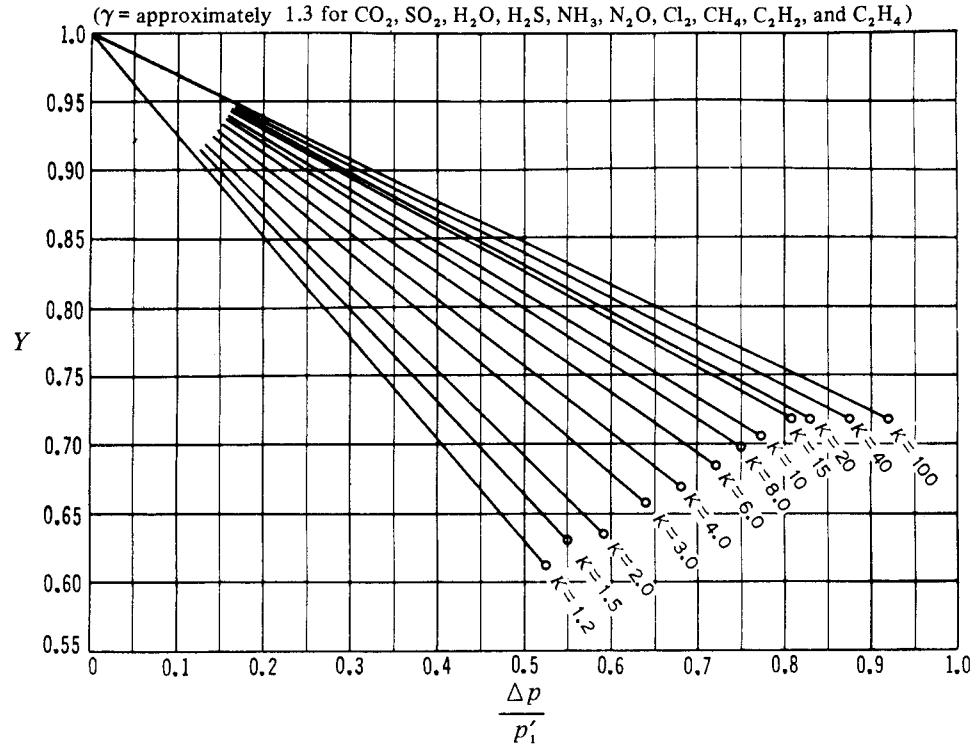


$$\frac{\Delta p}{p'_1} = \frac{\Delta P}{P'_1} \qquad \frac{p'_2}{p'_1} = \frac{P'_2}{P'_1}$$

Net Expansion Factor Y for Compressible Flow Through Pipe to a Larger Flow Area

$\gamma = 1.3$

(γ = approximately 1.3 for CO_2, SO_2, H_2O, H_2S, NH_3, N_2O, Cl_2, CH_4, C_2H_2, and C_2H_4)



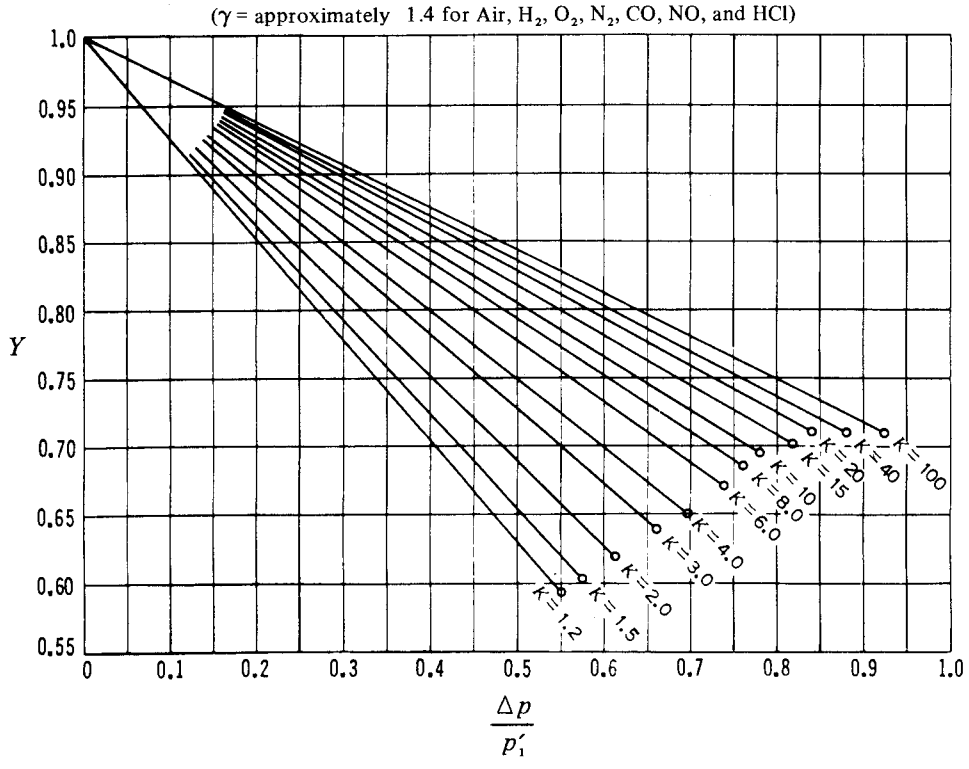
Limiting Factors For Sonic Velocity

$\gamma = 1.3$

K	$\frac{\Delta p}{p'_1}$	Y
1.2	.525	.612
1.5	.550	.631
2.0	.593	.635
3	.642	.658
4	.678	.670
6	.722	.685
8	.750	.698
10	.773	.705
15	.807	.718
20	.831	.718
40	.877	.718
100	.920	.718

$\gamma = 1.4$

(γ = approximately 1.4 for Air, H_2, O_2, N_2, CO, NO, and HCl)



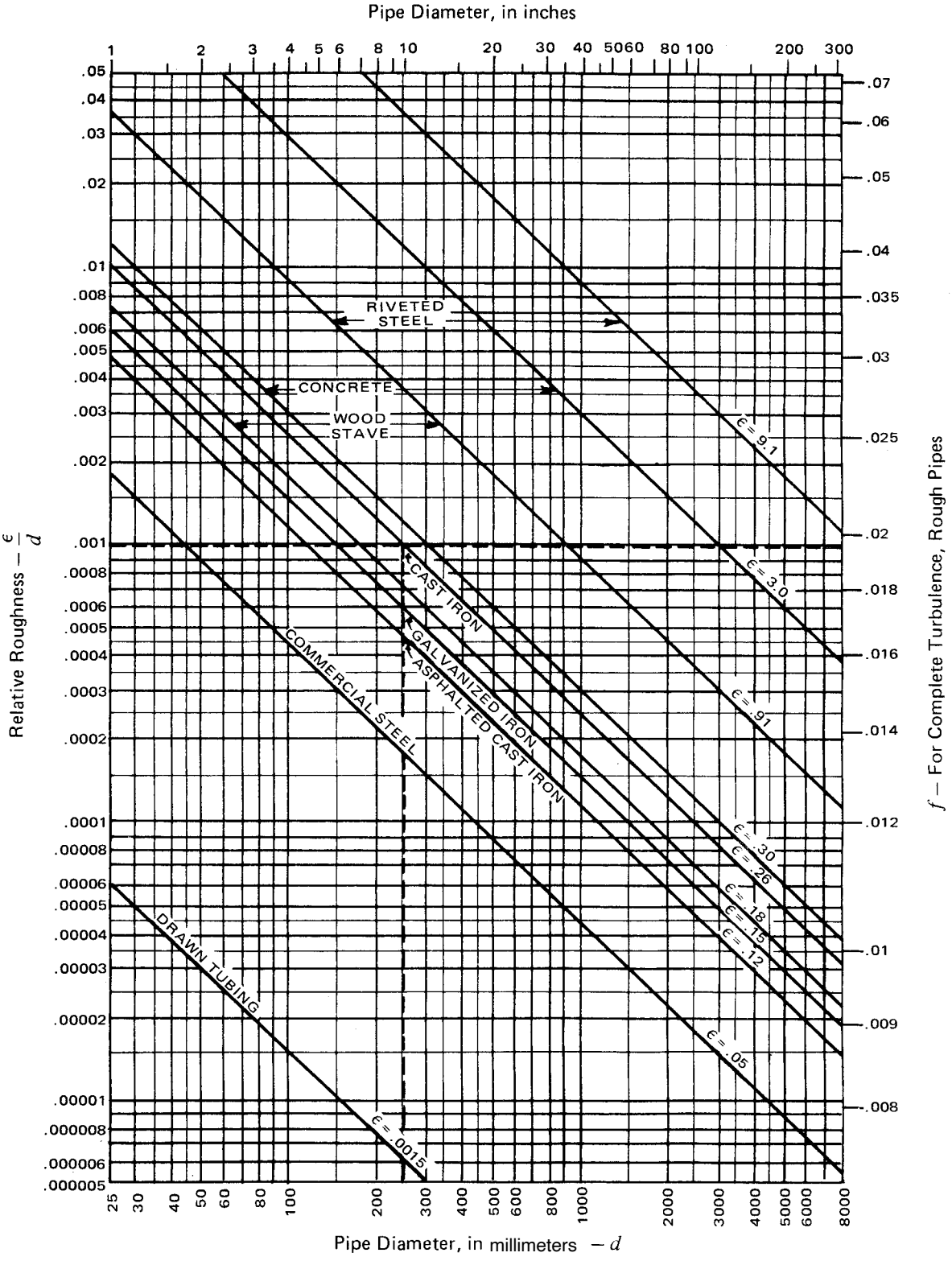
Limiting Factors For Sonic Velocity

$\gamma = 1.4$

K	$\frac{\Delta p}{p'_1}$	Y
1.2	.552	.588
1.5	.576	.606
2.0	.612	.622
3	.662	.639
4	.697	.649
6	.737	.671
8	.762	.685
10	.784	.695
15	.818	.702
20	.839	.710
40	.883	.710
100	.926	.710

$$\frac{\Delta p}{p'_1} = \frac{\Delta P}{P'_1}$$

Relative Roughness of Pipe Materials and Friction Factors For Complete Turbulence

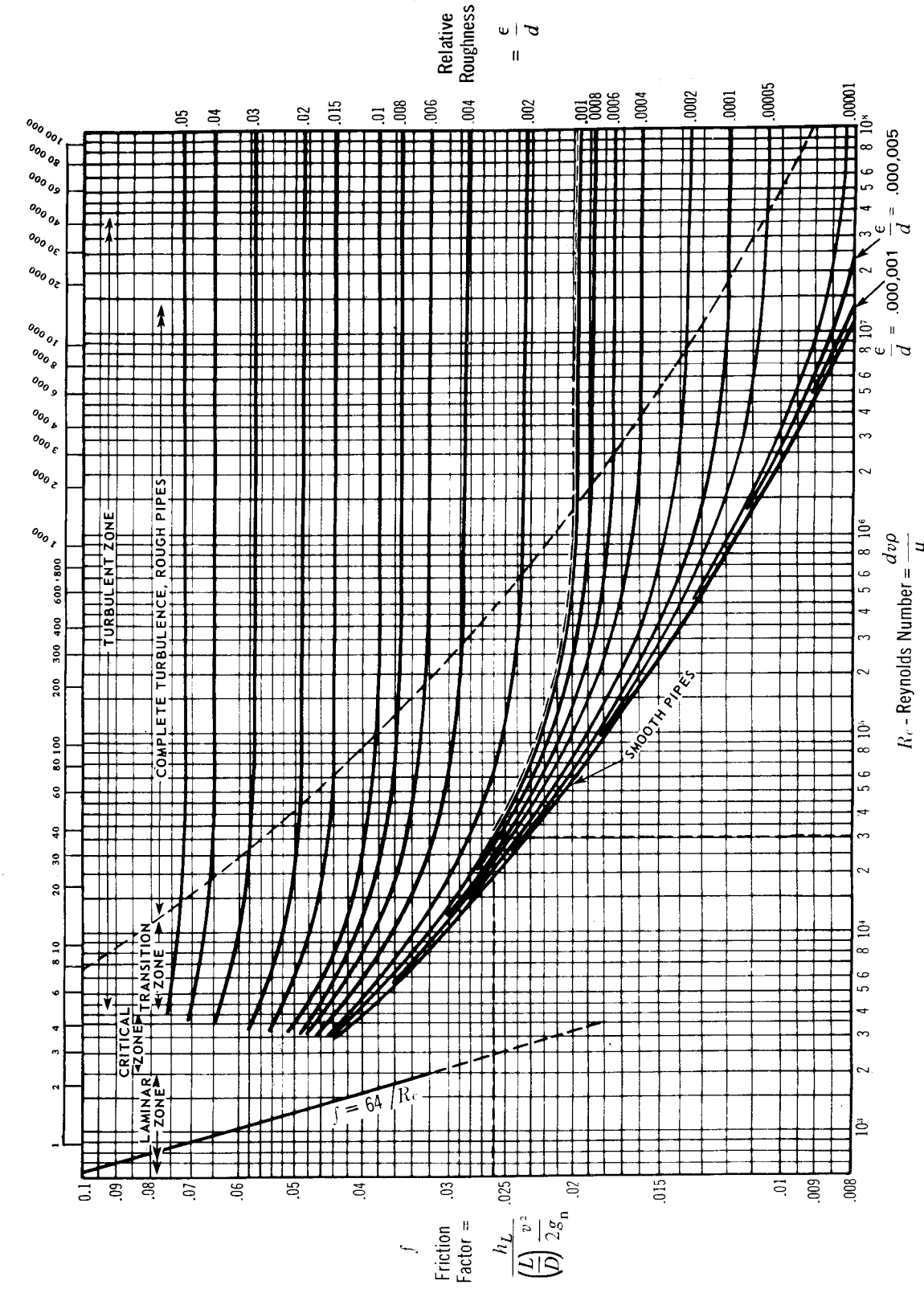


(Absolute Roughness ϵ is in millimeters)

Problem: Determine absolute and relative roughness, and friction factor, for fully turbulent flow in a cast iron pipe, 250 mm int. diam.

Solution: Absolute roughness (ϵ) = 0.26 Relative roughness (ϵ/d) = 0.001

Friction Factors for Any Type of Commercial Pipe



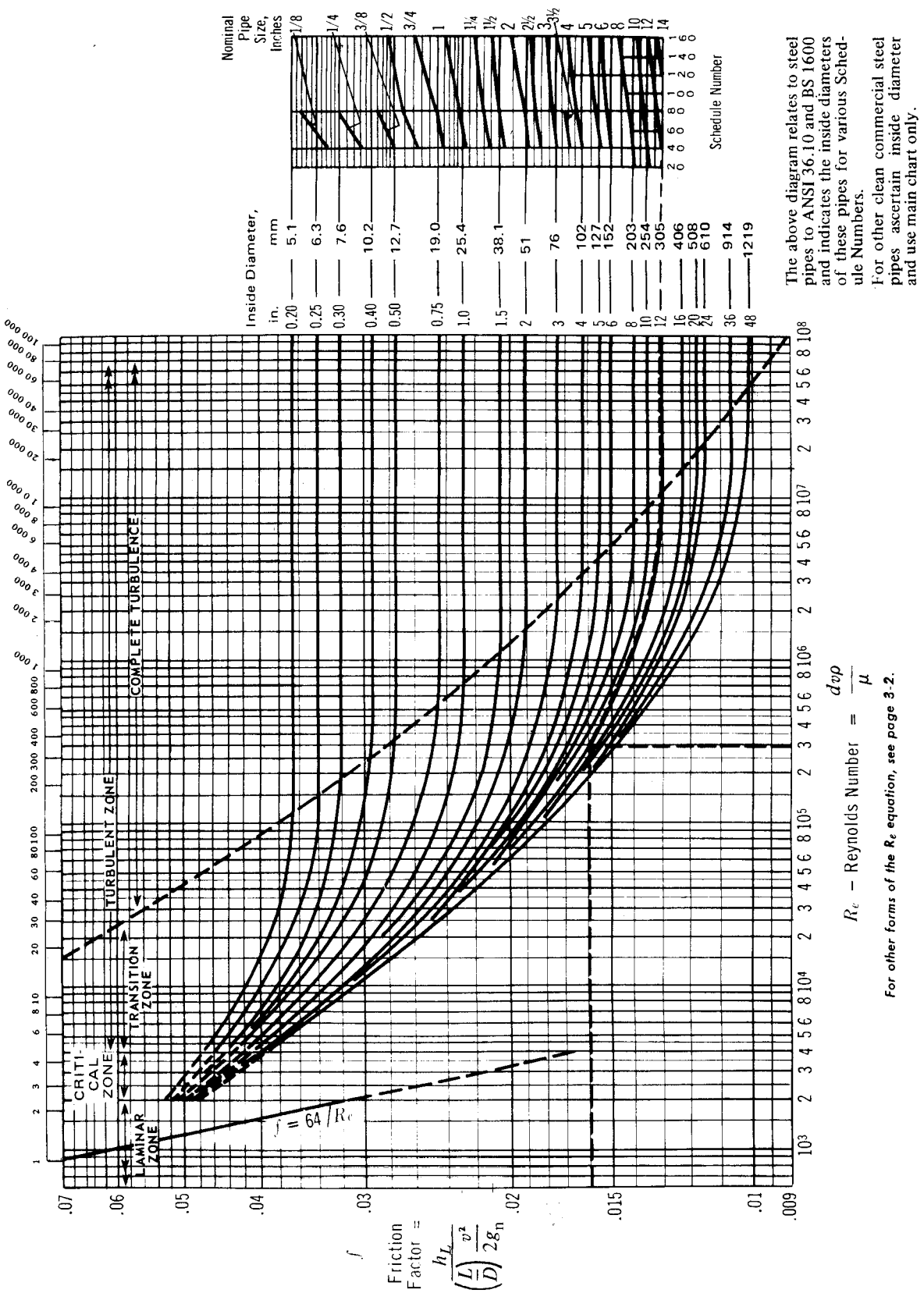
Problem: Determine the friction factor for a cast iron pipe 250 mm int. diam. at a Reynolds number flow of 30,000.

Solution: The relative roughness (see page A-23) is 0.001. Then, the friction factor (f) equals 0.026.

For other forms of the R_e equation, see page 3-2.

Adapted from data extracted from Bibliography reference 18.

Friction Factors for Clean Commercial Steel Pipes



The above diagram relates to steel pipes to ANSI 36.10 and BS 1600 and indicates the inside diameters of these pipes for various Schedule Numbers.

For other clean commercial steel pipes ascertain inside diameter and use main chart only.

For other forms of the R_e equation, see page 3-2.

Problem: Determine the friction factor for 12-inch Schedule 40 pipe at a flow having a Reynolds number of 300 000.

Solution: The friction factor (f) equals 0.016.

Adapted from data extracted from Bibliography reference 18.

"K" FACTOR TABLE – SHEET 1 of 4

Representative Resistance Coefficients (K) for Valves and Fittings

("K" is based on use of schedule pipe as listed on page 2-10)

PIPE FRICTION DATA FOR CLEAN COMMERCIAL STEEL PIPE WITH FLOW IN ZONE OF COMPLETE TURBULENCE

Nominal Size	mm	15	20	25	32	40	50	65, 80	100	125	150	200, 250	300-400	450-600
	in.	½	¾	1	1¼	1½	2	2½, 3	4	5	6	8, 10	12-16	18-24
Friction Factor (f_T)		.027	.025	.023	.022	.021	.019	.018	.017	.016	.015	.014	.013	.012

FORMULAS FOR CALCULATING "K" FACTORS * FOR VALVES AND FITTINGS WITH REDUCED PORT
(Ref: Pages 2-11 and 3-4)

Formula 1

$$K_2 = \frac{0.8 \sin\frac{\theta}{2}(1-\beta^2)}{\beta^4} = \frac{K_1}{\beta^4}$$

Formula 2

$$K_2 = \frac{0.5\,(1-\beta^2)\sqrt{\sin\frac{\theta}{2}}}{\beta^4} = \frac{K_1}{\beta^4}$$

Formula 3

$$K_2 = \frac{2.6 \sin\frac{\theta}{2}(1-\beta^2)^2}{\beta^4} = \frac{K_1}{\beta^4}$$

Formula 4

$$K_2 = \frac{(1-\beta^2)^2}{\beta^4} = \frac{K_1}{\beta^4}$$

Formula 5

$$K_2 = \frac{K_1}{\beta^4} + \text{Formula 1} + \text{Formula 3}$$

$$K_2 = \frac{K_1 + \sin\frac{\theta}{2}\left[0.8\,(1-\beta^2) + 2.6\,(1-\beta^2)^2\right]}{\beta^4}$$

Formula 6

$$K_2 = \frac{K_1}{\beta^4} + \text{Formula 2} + \text{Formula 4}$$

$$K_2 = \frac{K_1 + 0.5\sqrt{\sin\frac{\theta}{2}}(1-\beta^2) + (1-\beta^2)^2}{\beta^4}$$

Formula 7

$$K_2 = \frac{K_1}{\beta^4} + \beta\,(\text{Formula 2} + \text{Formula 4}) \text{ when } \theta = 180°$$

$$K_2 = \frac{K_1 + \beta\,[0.5\,(1-\beta^2) + (1-\beta^2)^2]}{\beta^4}$$

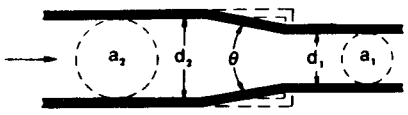
$$\beta = \frac{d_1}{d_2}$$

$$\beta^2 = \left(\frac{d_1}{d_2}\right)^2 = \frac{a_1}{a_2}$$

Subscript 1 defines dimensions and coefficients with reference to the smaller diameter.
Subscript 2 refers to the larger diameter.

***Use K furnished by valve or fitting supplier when available**

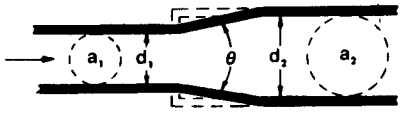
SUDDEN AND GRADUAL CONTRACTION



If: $\theta \leq 45°$ ..... K_2 = Formula 1

$45° < \theta \leq 180°$ .... K_2 = Formula 2

SUDDEN AND GRADUAL ENLARGEMENT



If: $\theta \leq 45°$ ..... K_2 = Formula 3

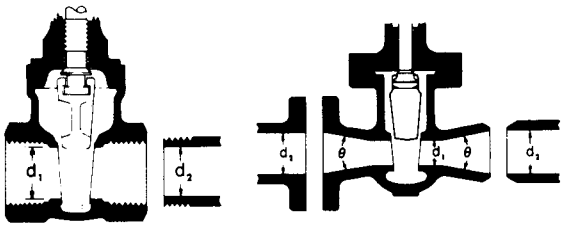
$45° < \theta \leq 180°$ .... K_2 = Formula 4

"K" FACTOR TABLE – SHEET 2 of 4

Representative Resistance Coefficients (K) for Valves and Fittings

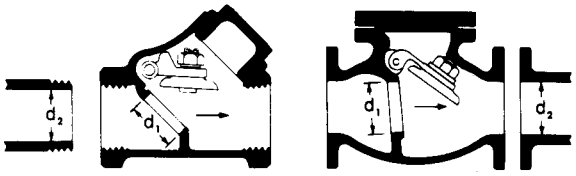
(for formulas and friction data, see page A-26)
("K" is based on use of schedule pipe as listed on page 2-10)

GATE VALVES
Wedge Disc, Double Disc, or Plug Type



If: $\beta = 1, \theta = 0$ ..... $K_1 = 8 f_T$
$\beta < 1$ and $\theta \leq 45°$ ..... K_2 = Formula 5
$\beta < 1$ and $45° < \theta \leq 180°$ .... K_2 = Formula 6

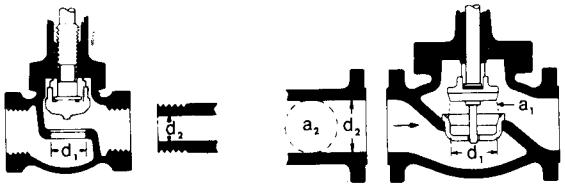
SWING CHECK VALVES



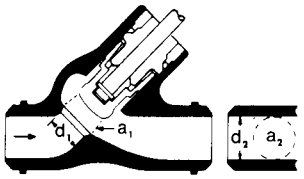
$K = 100 f_T$ $K = 50 f_T$

Minimum pipe velocity (mps) for full disc lift
$= 45\sqrt{\bar{V}}$ $= 75\sqrt{\bar{V}}$ except
U/L listed $= 120\sqrt{\bar{V}}$

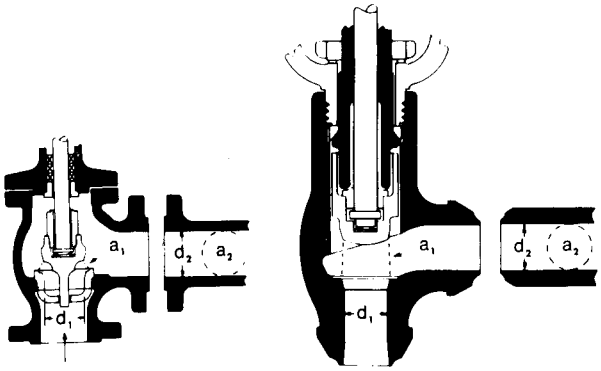
GLOBE AND ANGLE VALVES



If: $\beta = 1$ $K_1 = 340 f_t$



If: $\beta = 1 \ldots . K_1 = 55 f_T$

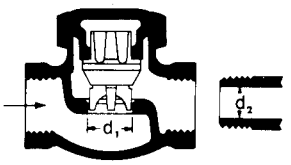


If: $\beta = 1 \ldots K_1 = 150 f_T$ If: $\beta = 1 \ldots . K_1 = 55 f_T$

All globe and angle valves,
whether reduced seat or throttled,

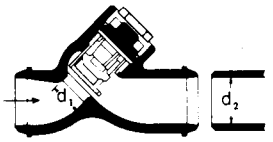
If: $\beta < 1 \ldots . K_2$ = Formula 7

LIFT CHECK VALVES



If: $\beta = 1 \ldots . K_1 = 600 f_T$
$\beta < 1 \ldots . K_2$ = Formula 7

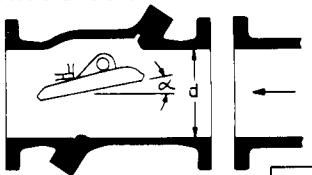
Minimum pipe velocity (mps) for full disc lift
$= 50 \beta^2 \sqrt{\bar{V}}$



If: $\beta = 1 \ldots . K_1 = 55 f_T$
$\beta < 1 \ldots . K_2$ = Formula 7

Minimum pipe velocity (mps) for full disc lift
$= 170 \beta^2 \sqrt{\bar{V}}$

TILTING DISC CHECK VALVES



Sizes	$\alpha = 5°$	$\alpha = 15°$
50 mm (2") to 200 mm (8") $K =$	$40 f_T$	$120 f_T$
250 mm (10") to 350 mm (14") $K =$	$30 f_T$	$90 f_T$
400 mm (16") to 1200 mm (48") $K =$	$20 f_T$	$60 f_T$
Minimum pipe velocity (mps) for full disc lift =	$100\sqrt{\bar{V}}$	$40\sqrt{\bar{V}}$

Note. mps = metres per second

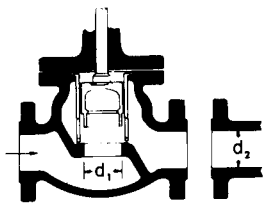
"K" FACTOR TABLE – SHEET 3 of 4

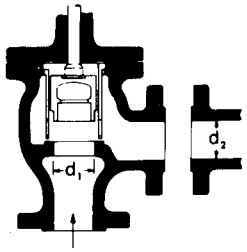
Representative Resistance Coefficients (K) for Valves and Fittings

(for formulas and friction data, see page A-26)

("K" is based on use of schedule type as listed on page 2-10)

STOP-CHECK VALVES (Globe and Angle Types)





If:

$\beta = 1 \ldots . \ K_1 = 400 f_T$

$\beta < 1 \ldots . \ K_2 =$ Formula 7

Minimum pipe velocity (mps) for full disc lift

$= 70 \beta^2 \sqrt{\bar{V}}$

If:

$\beta = 1 \ldots . \ K_1 = 200 f_T$

$\beta < 1 \ldots . \ K_2 =$ Formula 7

Minimum pipe velocity (mps) for full disc lift

$= 95 \beta^2 \sqrt{\bar{V}}$

FOOT VALVES WITH STRAINER

Poppet Disc

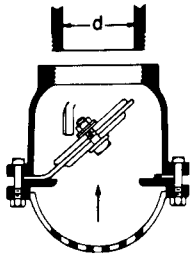


$K = 420 f_T$

Minimum pipe velocity (mps) for full disc lift

$= 20 \sqrt{\bar{V}}$

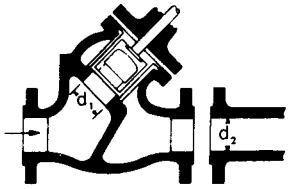
Hinged Disc

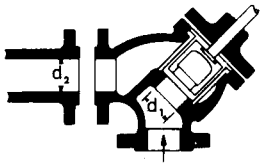


$K = 75 f_T$

Minimum pipe velocity (mps) for full disc lift

$= 45 \sqrt{\bar{V}}$





If:

$\beta = 1 \ldots . \ K_1 = 300 f_T$

$\beta < 1 \ldots . \ K_2 =$ Formula 7

If:

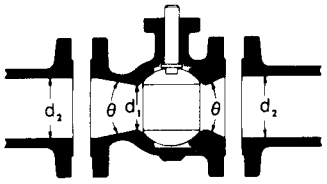
$\beta = 1 \ldots . \ K_1 = 350 f_T$

$\beta < 1 \ldots . \ K_2 =$ Formula 7

Minimum pipe velocity (mps) for full disc lift

$= 75 \beta^2 \sqrt{\bar{V}}$

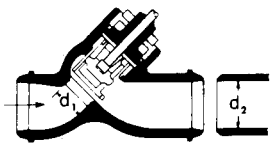
BALL VALVES

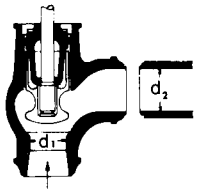


If: $\beta = 1, \theta = 0 \ldots . \ K_1 = 3 f_T$

$\beta < 1$ and $\theta \leq 45° \ldots . \ K_2 =$ Formula 5

$\beta < 1$ and $45° < \theta \leq 180° \ldots . \ K_2 =$ Formula 6





$\beta = 1 \ldots . \ K_1 = 55 f_T$

$\beta < 1 \ldots . \ K_2 =$ Formula 7

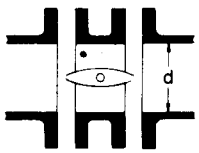
$\beta = 1 \ldots . \ K_1 = 55 f_T$

$\beta < 1 \ldots . \ K_2 =$ Formula 7

Minimum pipe velocity (mps) for full disc lift

$= 170 \beta^2 \sqrt{\bar{V}}$

BUTTERFLY VALVES



Sizes 50 mm (2″) to 200 mm (8″) . . . . . $K = 45 f_T$

Sizes 250 mm (10″) to 350 mm (14″) . . . . . $K = 35 f_T$

Sizes 400 mm (16″) to 600 mm (24″) . . . . . $K = 25 f_T$

"K" FACTOR TABLE – SHEET 4 of 4

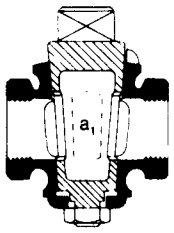
Representative Resistance Coefficients (K) for Valves and Fittings

(for formulas and friction data, see page A-26)

("K" is based on use of schedule pipe as listed on page 2-10)

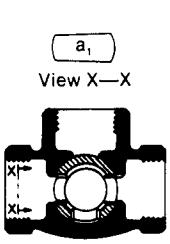
PLUG VALVES AND COCKS

Straight-Way

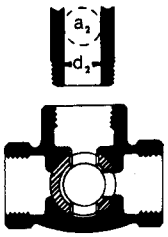


If: $\beta = 1$, $K_1 = 18\, f_T$

3-Way



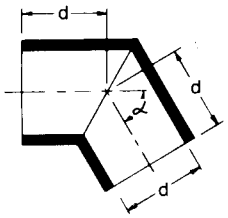
If: $\beta = 1$, $K_1 = 30\, f_T$



If: $\beta = 1$, $K_1 = 90\, f_T$

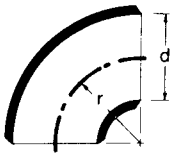
If: $\beta < 1$ K_2 = Formula 6

MITRE BENDS



α	K
0°	$2\, f_T$
15°	$4\, f_T$
30°	$8\, f_T$
45°	$15\, f_T$
60°	$25\, f_T$
75°	$40\, f_T$
90°	$60\, f_T$

90° PIPE BENDS AND FLANGED OR BUTT-WELDING 90° ELBOWS



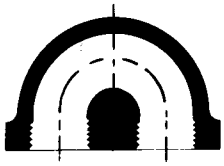
r/d	K	r/d	K
1	$20\, f_T$	8	$24\, f_T$
1.5	$14\, f_T$	10	$30\, f_T$
2	$12\, f_T$	12	$34\, f_T$
3	$12\, f_T$	14	$38\, f_T$
4	$14\, f_T$	16	$42\, f_T$
6	$17\, f_T$	20	$50\, f_T$

The resistance coefficient, K_B, for pipe bends other than 90° may be determined as follows:

$$K_B = (n-1)\left(0.25\,\pi\, f_T \frac{r}{d} + 0.5\,K\right) + K$$

n = number of 90° bends
K = resistance coefficient for one 90° bend (per table)

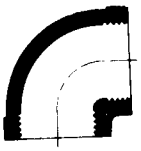
CLOSE PATTERN RETURN BENDS



$K = 50\, f_T$

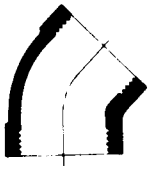
STANDARD ELBOWS

90°



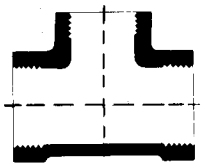
$K = 30\, f_T$

45°



$K = 16\, f_T$

STANDARD TEES

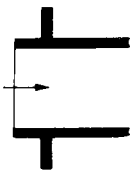


Flow thru run . . . . . $K = 20\, f_T$

Flow thru branch . . . . . $K = 60\, f_T$

PIPE ENTRANCE

Inward Projecting

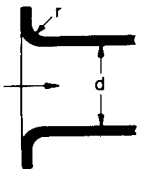


$K = 0.78$

r/d	K
0.00*	0.5
0.02	0.28
0.04	0.24
0.06	0.15
0.10	0.09
0.15 & up	0.04

*Sharp-edged

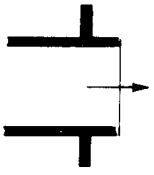
Flush



For K, see table

PIPE EXIT

Projecting



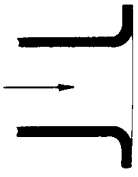
$K = 1.0$

Sharp-Edged



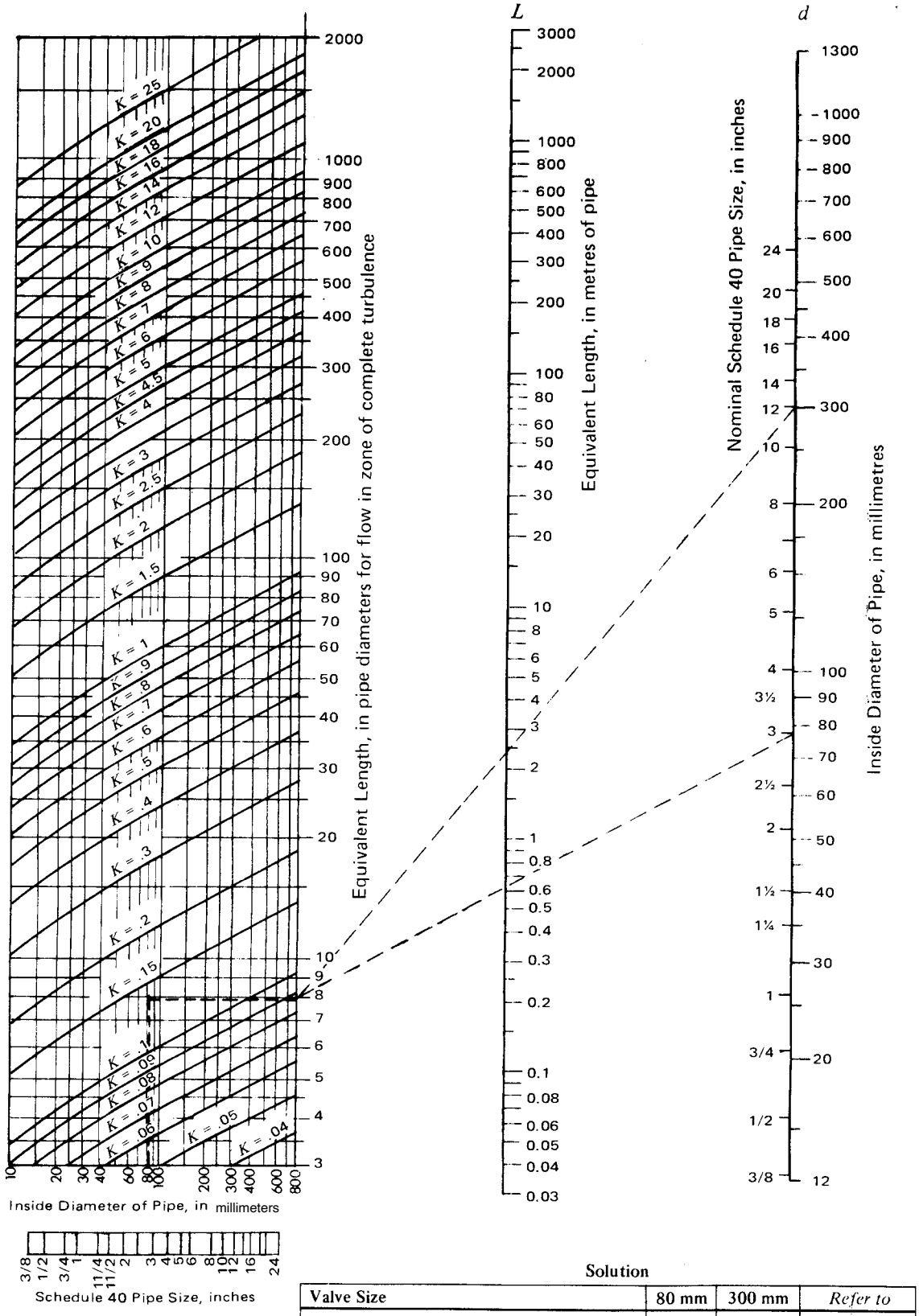
$K = 1.0$

Rounded



$K = 1.0$

Equivalent Lengths *L* and *L/D* and Resistance Coefficient *K*



Problem: Find the equivalent length in pipe diameters and metres of Schedule 40 clean commercial steel pipe, and the resistance factor *K* for 80 mm and 300 mm fully opened gate valves, ANSI Class 300, with flow in zone of complete turbulence.

Solution

Valve Size	80 mm	300 mm	*Refer to*
Equivalent length, pipe diameters	8	8	Page A-27
Equivalent length, metres of Sched. 40 pipe	0.62	2.43	Dotted lines
Resist Factor *K*, based on Sched. 40 pipe	0.14	0.10	on chart

For discussion on *L/D* and *K* see pages 2-8 to 2-10